

Derivations and the Adjoint Action

Road map

The adjoint construction lets elements of a Lie algebra act as linear operators on the Lie algebra itself. We begin with the physics hierarchy of states, operators, and superoperators. We then define derivations and prove in full detail that they form a Lie subalgebra of $\mathfrak{gl}(\mathfrak{g})$. The Jacobi identity produces a distinguished family of derivations, $\text{ad}_X(Y) = [X, Y]$, called inner derivations. We prove

$$[\text{ad}_X, \text{ad}_Y] = \text{ad}_{[X, Y]},$$

identify the kernel of ad with the centre of $\mathfrak{g}$, and deduce $\mathfrak{g}/Z(\mathfrak{g}) \cong \text{ider}(\mathfrak{g})$. Finally, we relate the infinitesimal map ad to the group adjoint action Ad .

16.1 From operators to superoperators

In quantum mechanics, a Hilbert space V is the space of state vectors. At this stage V is simply a vector space with an inner product; it is not itself a Lie algebra. A linear operator is a linear map

$$X : V \longrightarrow V.$$

The collection of all such operators is

$$\mathfrak{gl}(V) := \text{End}(V) = \{X : V \rightarrow V : X \text{ is linear}\}.$$

Composition turns $\text{End}(V)$ into an associative algebra. Forgetting the associative product but retaining its commutator gives the operator Lie algebra

$$[X_1, X_2] := X_1 \circ X_2 - X_2 \circ X_1. \quad (16.1)$$

We shall write

$$\mathfrak{g} := \mathfrak{gl}(V)$$

in this motivating example.

For a fixed $X \in \mathfrak{g}$, the rule

$$Y \longmapsto [X, Y]$$

is linear in Y . It is therefore a linear map from $\mathfrak{g}$ to itself. Such maps act on operators rather than directly on state vectors and are consequently called *superoperators* in physics.

Definition 16.1: The superoperator algebra

For any Lie algebra $\mathfrak{g}$,

$$\mathfrak{gl}(\mathfrak{g}) := \text{End}(\mathfrak{g}) = \{\mathcal{Y} : \mathfrak{g} \rightarrow \mathfrak{g} : \mathcal{Y} \text{ is linear}\}.$$

It is itself a Lie algebra under the commutator

$$[\mathcal{Y}_1, \mathcal{Y}_2] := \mathcal{Y}_1 \circ \mathcal{Y}_2 - \mathcal{Y}_2 \circ \mathcal{Y}_1.$$

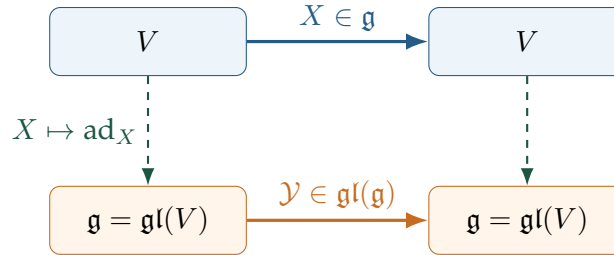


图 16.1: An operator acts on states, whereas a superoperator acts linearly on operators. The adjoint construction sends an operator X to the superoperator $Y \mapsto [X, Y]$.

Thus $\mathfrak{gl}(\mathfrak{g})$ constructs new operators linearly from old ones. The central example will be the operation

$$Y \mapsto [X, Y],$$

which turns an old operator Y into the commutator with a fixed X .

16.2 Derivations of a Lie algebra

Definition 16.2: Derivation

Let $\mathfrak{g}$ be a Lie algebra over a field $\mathbb{K}$. A linear map

$$D : \mathfrak{g} \longrightarrow \mathfrak{g}$$

is a *derivation* if it obeys the Leibniz rule for the Lie bracket:

$$D([X, Y]) = [D(X), Y] + [X, D(Y)] \quad (X, Y \in \mathfrak{g}). \quad (16.2)$$

The vector space of all derivations is denoted

$$\text{Der}(\mathfrak{g}) := \{D \in \mathfrak{gl}(\mathfrak{g}) : D \text{ satisfies (16.2)}\}.$$

Equation (16.2) is the product rule adapted to the Lie bracket. It should not be confused with the homomorphism condition

$$D([X, Y]) = [D(X), D(Y)].$$

A derivation differentiates a bracket; it does not usually preserve the bracket as a Lie-algebra homomorphism.

Theorem 16.1: Derivations form a Lie algebra

For every Lie algebra $\mathfrak{g}$,

$$\text{Der}(\mathfrak{g}) \leq \mathfrak{gl}(\mathfrak{g})$$

is a Lie subalgebra. Its bracket is the operator commutator

$$[D_1, D_2] = D_1 D_2 - D_2 D_1.$$

证明. We check vector-space closure and bracket closure separately.

Linear combinations. Let $D_1, D_2 \in \text{Der}(\mathfrak{g})$ and $\alpha_1, \alpha_2 \in \mathbb{K}$. Then

$$\begin{aligned} (\alpha_1 D_1 + \alpha_2 D_2)([X, Y]) &= \alpha_1 D_1([X, Y]) + \alpha_2 D_2([X, Y]) \\ &= \alpha_1 [D_1 X, Y] + \alpha_1 [X, D_1 Y] \\ &\quad + \alpha_2 [D_2 X, Y] + \alpha_2 [X, D_2 Y] \\ &= [(\alpha_1 D_1 + \alpha_2 D_2)X, Y] \\ &\quad + [X, (\alpha_1 D_1 + \alpha_2 D_2)Y]. \end{aligned}$$

Thus $\alpha_1 D_1 + \alpha_2 D_2$ is again a derivation.

Commutators. For $D_1, D_2 \in \text{Der}(\mathfrak{g})$, expand both compositions using the derivation rule:

$$\begin{aligned} [D_1, D_2]([X, Y]) &= D_1 D_2([X, Y]) - D_2 D_1([X, Y]) \\ &= D_1([D_2 X, Y] + [X, D_2 Y]) \\ &\quad - D_2([D_1 X, Y] + [X, D_1 Y]) \\ &= [D_1 D_2 X, Y] + [D_2 X, D_1 Y] \\ &\quad + [D_1 X, D_2 Y] + [X, D_1 D_2 Y] \\ &\quad - [D_2 D_1 X, Y] - [D_1 X, D_2 Y] \\ &\quad - [D_2 X, D_1 Y] - [X, D_2 D_1 Y]. \end{aligned}$$

The two mixed pairs cancel. What remains is

$$\begin{aligned} [D_1, D_2]([X, Y]) &= [(D_1 D_2 - D_2 D_1)X, Y] \\ &\quad + [X, (D_1 D_2 - D_2 D_1)Y] \\ &= [[D_1, D_2]X, Y] + [X, [D_1, D_2]Y]. \end{aligned}$$

Hence $[D_1, D_2]$ is a derivation. This proves that $\text{Der}(\mathfrak{g})$ is a Lie subalgebra. ■

16.3 Inner derivations

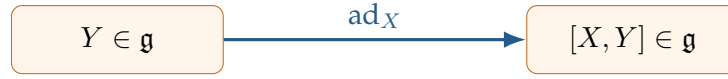
Every Lie algebra comes with a distinguished family of derivations obtained from its own elements.

Definition 16.3: Infinitesimal adjoint action

For $X \in \mathfrak{g}$, define

$$\mathrm{ad}_X : \mathfrak{g} \longrightarrow \mathfrak{g}, \quad \mathrm{ad}_X(Y) := [X, Y]. \quad (16.3)$$

This map is called the *adjoint action of X* or the *infinitesimal adjoint action*.



the fixed element X acts on all of $\mathfrak{g}$

图 16.2: The adjoint action of a fixed X sends Y to the bracket $[X, Y]$.

Proposition 16.1: Every adjoint action is a derivation

For every $X \in \mathfrak{g}$,

$$\mathrm{ad}_X \in \mathrm{Der}(\mathfrak{g}).$$

证明. Let $Y, Z \in \mathfrak{g}$. The Jacobi identity gives

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

Using antisymmetry twice,

$$[Y, [Z, X]] = -[Y, [X, Z]], \quad [Z, [X, Y]] = -[[X, Y], Z].$$

Therefore

$$\begin{aligned} \mathrm{ad}_X([Y, Z]) &= [X, [Y, Z]] \\ &= [Y, [X, Z]] + [[X, Y], Z] \\ &= [Y, \mathrm{ad}_X Z] + [\mathrm{ad}_X Y, Z] \\ &= [\mathrm{ad}_X Y, Z] + [Y, \mathrm{ad}_X Z]. \end{aligned}$$

This is exactly the derivation rule. ■

Definition 16.4: Inner and outer derivations

The set

$$\mathrm{id}(\mathfrak{g}) := \{\mathrm{ad}_X : X \in \mathfrak{g}\} \quad (16.4)$$

is the set of *inner derivations*. A derivation not belonging to this set is called *outer*. Once we prove that $\mathrm{id}(\mathfrak{g})$ is an ideal in $\mathrm{Der}(\mathfrak{g})$, the quotient

$$\mathrm{Out}(\mathfrak{g}) := \mathrm{Der}(\mathfrak{g}) / \mathrm{id}(\mathfrak{g})$$

will be the Lie algebra of outer-derivation classes.

Proposition 16.2: Bracket of inner derivations

For every $X, Y \in \mathfrak{g}$,

$$[\mathrm{ad}_X, \mathrm{ad}_Y] = \mathrm{ad}_{[X, Y]}. \quad (16.5)$$

Consequently,

$$\mathrm{id}(\mathfrak{g}) \leq \mathrm{Der}(\mathfrak{g})$$

is a Lie subalgebra.

证明. Apply the left-hand side to an arbitrary $Z \in \mathfrak{g}$:

$$\begin{aligned} [\mathrm{ad}_X, \mathrm{ad}_Y](Z) &= \mathrm{ad}_X(\mathrm{ad}_Y Z) - \mathrm{ad}_Y(\mathrm{ad}_X Z) \\ &= [X, [Y, Z]] - [Y, [X, Z]] \\ &= [X, [Y, Z]] + [Y, [Z, X]] \\ &= -[Z, [X, Y]] \quad \text{by the Jacobi identity} \\ &= [[X, Y], Z] \\ &= \mathrm{ad}_{[X, Y]}(Z). \end{aligned}$$

Since the two linear maps agree on every Z , they are equal. ■

Proposition 16.3: Inner derivations form an ideal

For $D \in \mathrm{Der}(\mathfrak{g})$ and $X \in \mathfrak{g}$,

$$[D, \mathrm{ad}_X] = \mathrm{ad}_{D(X)}. \quad (16.6)$$

Hence $\mathrm{id}(\mathfrak{g}) \triangleleft \mathrm{Der}(\mathfrak{g})$.

证明. For $Y \in \mathfrak{g}$,

$$\begin{aligned} [D, \mathrm{ad}_X](Y) &= D([X, Y]) - [X, D(Y)] \\ &= [D(X), Y] + [X, D(Y)] - [X, D(Y)] \\ &= [D(X), Y] \\ &= \mathrm{ad}_{D(X)}(Y). \end{aligned}$$
■

16.4 The adjoint representation and the centre

Theorem 16.2: The adjoint representation of a Lie algebra

The map

$$\mathrm{ad} : \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathfrak{g}), \quad X \longmapsto \mathrm{ad}_X, \quad (16.7)$$

is a Lie-algebra homomorphism. Its image and kernel are

$$\mathrm{im}(\mathrm{ad}) = \mathrm{id}(\mathfrak{g}), \quad \ker(\mathrm{ad}) = Z(\mathfrak{g}). \quad (16.8)$$

证明. Linearity follows directly from bilinearity of the bracket:

$$\begin{aligned}\mathrm{ad}_{\alpha X + \beta Y}(Z) &= [\alpha X + \beta Y, Z] \\ &= \alpha[X, Z] + \beta[Y, Z] \\ &= (\alpha \mathrm{ad}_X + \beta \mathrm{ad}_Y)(Z).\end{aligned}$$

Identity (16.5) says precisely that brackets are preserved:

$$\mathrm{ad}_{[X, Y]} = [\mathrm{ad}_X, \mathrm{ad}_Y].$$

Thus ad is a Lie-algebra homomorphism.

Its image is the set of all maps ad_X , which is $\mathrm{ider}(\mathfrak{g})$ by definition. Finally,

$$\begin{aligned}\ker(\mathrm{ad}) &= \{X \in \mathfrak{g} : \mathrm{ad}_X = 0\} \\ &= \{X \in \mathfrak{g} : [X, Y] = 0 \text{ for every } Y \in \mathfrak{g}\} \\ &= Z(\mathfrak{g}),\end{aligned}$$

the centre of $\mathfrak{g}$. ■

Theorem 16.3: Inner derivations as a quotient

The first isomorphism theorem applied to the adjoint representation gives

$$\boxed{\mathfrak{g}/Z(\mathfrak{g}) \cong \mathrm{ider}(\mathfrak{g})}. \quad (16.9)$$

In particular, if $\mathfrak{g}$ is centreless, then

$$\boxed{\mathfrak{g} \cong \mathrm{ider}(\mathfrak{g}), \quad X \longmapsto [X, \cdot]}. \quad (16.10)$$

证明. The homomorphism ad has kernel $Z(\mathfrak{g})$ and image $\mathrm{ider}(\mathfrak{g})$. The first isomorphism theorem gives

$$\mathfrak{g}/\ker(\mathrm{ad}) \cong \mathrm{im}(\mathrm{ad}),$$

which is (16.9). If $Z(\mathfrak{g}) = \{0\}$, then the quotient on the left is simply $\mathfrak{g}$. ■

16.5 The group adjoint action

For a linear Lie group G with $\mathfrak{g} = \mathrm{Lie}(G)$, conjugation by a group element preserves $\mathfrak{g}$, as proved in the preceding chapter.

Definition 16.5: Group adjoint action

For $A \in G$, define

$$\mathrm{Ad}_A : \mathfrak{g} \longrightarrow \mathfrak{g}, \quad \mathrm{Ad}_A(X) := AXA^{-1}. \quad (16.11)$$

The map

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}(\mathfrak{g}), \quad A \longmapsto \mathrm{Ad}_A$$

is called the *adjoint representation of the group*.

Proposition 16.4: The group adjoint map is a representationFor $A, B \in G$,

$$\mathrm{Ad}_{AB} = \mathrm{Ad}_A \circ \mathrm{Ad}_B. \quad (16.12)$$

Thus $\mathrm{Ad} : G \rightarrow \mathrm{GL}(\mathfrak{g})$ is a group homomorphism.证明. For $X \in \mathfrak{g}$,

$$\mathrm{Ad}_{AB}(X) = (AB)X(AB)^{-1} = A(BXB^{-1})A^{-1} = \mathrm{Ad}_A(\mathrm{Ad}_B X).$$

■

Theorem 16.4: Relation between Ad and ad For $X \in \mathfrak{g}$,

$$\mathrm{Ad}_{e^{tX}} = e^{t \mathrm{ad}_X} \quad (t \in \mathbb{R}). \quad (16.13)$$

Equivalently, the differential of the group representation Ad at the identity is the Lie-algebra representation ad :

$$\left. \frac{d}{dt} \mathrm{Ad}_{e^{tX}} \right|_{t=0} = \mathrm{ad}_X. \quad (16.14)$$

证明. Fix $Y \in \mathfrak{g}$ and put

$$F(t) := \mathrm{Ad}_{e^{tX}}(Y) = e^{tX} Y e^{-tX}.$$

Differentiate:

$$\begin{aligned} F'(t) &= X e^{tX} Y e^{-tX} - e^{tX} Y e^{-tX} X \\ &= [X, F(t)] \\ &= \mathrm{ad}_X(F(t)). \end{aligned}$$

The initial condition is $F(0) = Y$. The unique solution of the constant-coefficient linear differential equation

$$F'(t) = \mathrm{ad}_X F(t), \quad F(0) = Y,$$

is

$$F(t) = e^{t \mathrm{ad}_X} Y.$$

Since this holds for every Y , identity (16.13) follows. Differentiating it at $t = 0$ gives (16.14). ■

16.6 Examples

Example 16.1: An Abelian Lie algebra

If $\mathfrak{g}$ is Abelian, then every bracket is zero. Consequently

$$\mathrm{ad}_X = 0 \quad \text{for every } X,$$

so

$$\mathrm{id}(\mathfrak{g}) = \{0\}, \quad Z(\mathfrak{g}) = \mathfrak{g}.$$

On the other hand, the derivation rule becomes $D(0) = 0$, which every linear map satisfies. Hence

$$\mathrm{Der}(\mathfrak{g}) = \mathfrak{gl}(\mathfrak{g}).$$

Thus a nonzero linear endomorphism of a nonzero Abelian Lie algebra is an outer derivation.

Example 16.2: The full matrix algebra

Let $\mathfrak{g} = \mathfrak{gl}(V)$ with commutator bracket. Then

$$\mathrm{ad}_X(Y) = XY - YX.$$

The centre consists of the scalar operators:

$$Z(\mathfrak{gl}(V)) = \{\lambda I : \lambda \in \mathbb{K}\}$$

when V is finite-dimensional and nonzero. Therefore

$$\mathfrak{gl}(V)/\mathbb{K}I \cong \mathrm{id}(\mathfrak{gl}(V)).$$

Two matrices X and $X + \lambda I$ produce the same inner derivation, because the scalar matrix commutes with every operator.

Example 16.3: $\mathfrak{so}(3)$ and the cross product

Under the isomorphism $\mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$ from the preceding chapters,

$$\mathrm{ad}_x(y) = x \times y.$$

The only vector whose cross product with every y vanishes is $x = 0$. Thus $Z(\mathfrak{so}(3)) = \{0\}$, and

$$\mathfrak{so}(3) \cong \mathrm{id}(\mathfrak{so}(3)).$$

In coordinates,

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \longmapsto \mathrm{ad}_x = \begin{pmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{pmatrix},$$

because multiplication of this matrix by y produces $x \times y$.

What to retain from this chapter

1. $\mathfrak{gl}(\mathfrak{g})$ is the Lie algebra of linear superoperators $\mathfrak{g} \rightarrow \mathfrak{g}$, with commutator bracket.
2. A derivation satisfies the Lie-bracket product rule

$$D([X, Y]) = [DX, Y] + [X, DY].$$

The derivations form a Lie subalgebra

$$\text{Der}(\mathfrak{g}) \leq \mathfrak{gl}(\mathfrak{g}).$$

3. Every element $X \in \mathfrak{g}$ defines an inner derivation

$$\text{ad}_X(Y) = [X, Y].$$

The Jacobi identity is exactly what proves that this map is a derivation.

4. Inner derivations satisfy

$$[\text{ad}_X, \text{ad}_Y] = \text{ad}_{[X, Y]},$$

so $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is a Lie-algebra homomorphism.

- 5.

$$\text{im}(\text{ad}) = \text{ider}(\mathfrak{g}), \quad \ker(\text{ad}) = Z(\mathfrak{g}).$$

Hence

$$\mathfrak{g}/Z(\mathfrak{g}) \cong \text{ider}(\mathfrak{g}).$$

6. Inner derivations form an ideal in all derivations:

$$[D, \text{ad}_X] = \text{ad}_{D(X)}.$$

7. At group level,

$$\text{Ad}_A(X) = AXA^{-1}, \quad \text{Ad}_{AB} = \text{Ad}_A \text{Ad}_B.$$

The group and algebra adjoint actions are related by

$$\text{Ad}_{e^{tX}} = e^{t \text{ad}_X}.$$